

Apriori Association Rules

People Analytics · HR Risk

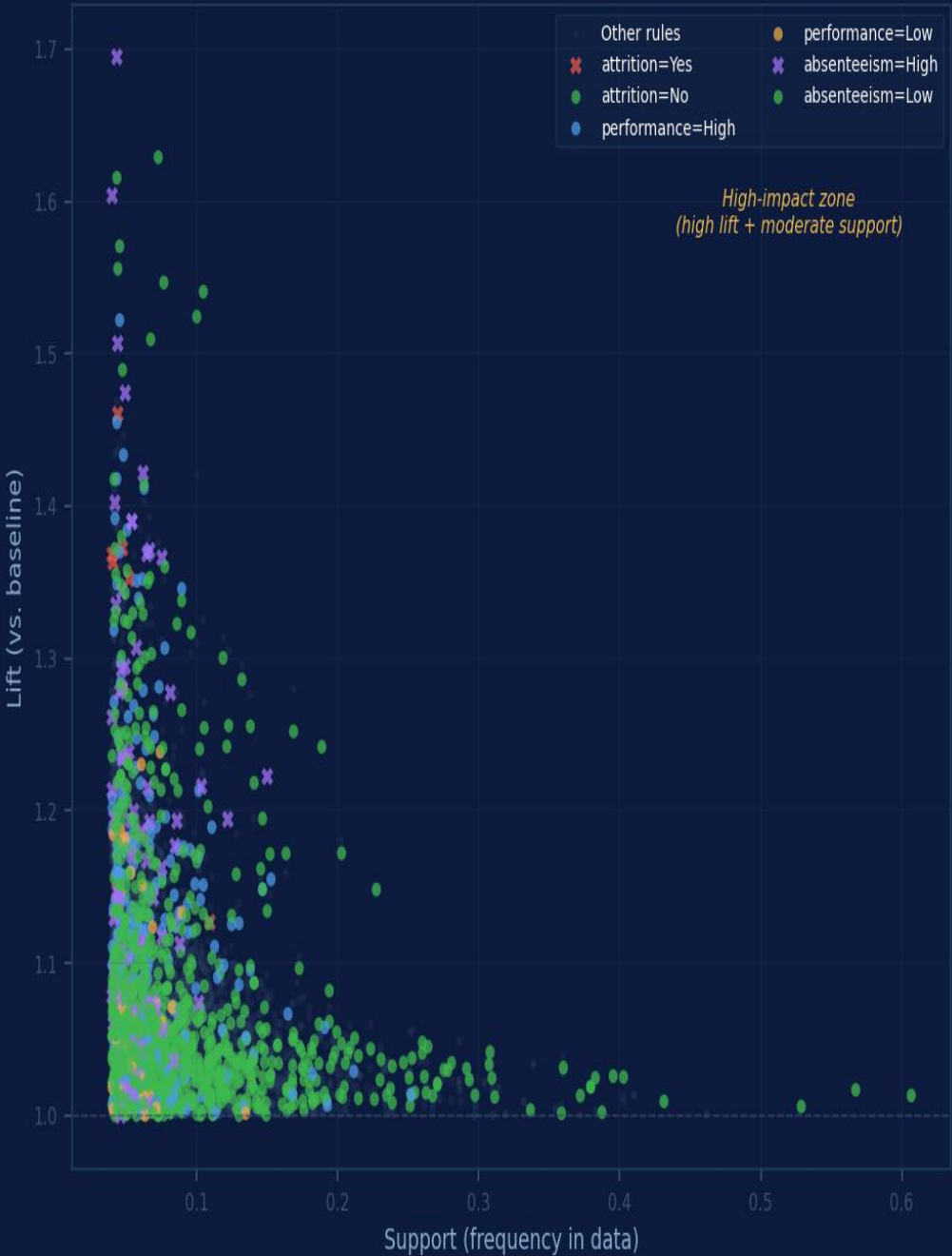
"15.2% leave.
That's a fact.
The policy question is: which profile?"

Apriori · k=4

min_sup=0.04

min_conf=0.60

Lift · Support



1,748

Employee Records

3,861

Frequent Itemsets

2,148

HR-Critical Rules

1.665

Max Lift (Absenteeism)



Averages Hide Structure

The quarterly report says attrition = 15.2%. The manager nods. Action item: improve engagement. Due date: Q4. The number is accurate. The action is directionless. The employees who leave are not a random 15.2% — they are a specific profile.



Combinations Are Non-Linear

night_shift alone: +12% attrition vs baseline.
no_remote alone: +9%. Together with no_training: lift = 1.480, confidence = 22.4%. The combination produces a risk that is not the sum of individual effects — it is the product of co-occurrence.



The Policy Gap

HR analytics produces reports. Apriori produces rules. A rule says: when {manager_rating=Low, remote_work=No} co-occur → absenteeism=High, lift=1.665. That rule names the intervention, the population, and the evidence level. A report does not.

Apriori does not predict individual risk. It discovers which HR combinations systematically produce which outcomes — at organizational scale.

HR DASHBOARD REPORTS

Night shift attrition: 17.2%

No-training attrition: 19.4%

Low manager attrition: 18.1%

→ *Numbers describe the problem*

→ *No conditions or combinations*

→ *Action: improve engagement*

→ *Owner: HR · Due: Q4*

→ *Same numbers next quarter*



APRIORI ASSOCIATION RULES

{manager_rating=Low, remote_work=No}

→ **absenteeism=High**

→ **Support: 4.2% · Lift: 1.665**

{remote_work=Yes, training_status=Completed,
manager_rating=High}

→ **attrition=No**

→ **Confidence: 100% · Lift: 1.175**

Each rule names:

- The antecedent profile (who)
- The outcome (what)
- The lift (how much above baseline)

OPS

HRIS / Workforce Management

department · shift · seniority

Organizational unit, work schedule (Day/Night), experience level (Junior/Mid/Senior) — structural HR dimensions

COMP

HR Administration / Payroll

contract_type · remote_work · overtime

Employment type (Permanent/Temporary), remote work access (Yes/No), regular overtime requirement — contract and flexibility dimensions

L&D

Learning Management System

training_status

Last training cycle completion status (Completed/Partial/None) — single most powerful individual predictor of attrition lift

PERF

Performance Management System

manager_rating · absenteeism · performance · attrition

Direct manager assessment (High/Med/Low) · Outcome variables included as items — all columns treated symmetrically in transaction encoding

Unlike every previous project: no StandardScaler. All 11 columns are categorical — TransactionEncoder converts each record into 30 binary items. Apriori was designed for this structure.

"Correlation is what dashboards show you. Association rules show you what conditions tend to occur together — and what outcome tends to follow. That's the difference between a report and a policy."

Rules, Not Scores

Every previous algorithm produces a score (IR), a label (–1), or a cluster (C3). Apriori produces an if-then rule with three metrics. A rule is directly translatable into an HR policy: IF these conditions exist, THEN this outcome is X times more likely — act on it.

Lift Beats Correlation

Lift = confidence / baseline. It answers: how much more likely is this outcome given this profile versus a randomly chosen employee? A rule with lift=1.665 is 66.5% above baseline. Correlation coefficients don't tell you that. Lift does.

Combinations Are the Signal

manager_rating=Low alone: lift=1.48 on attrition. remote_work=No alone: weaker effect. Together: lift=1.665 on absenteeism. The interaction term — the combination — is the policy-relevant signal that univariate reporting destroys by aggregation.

Purely Categorical by Design

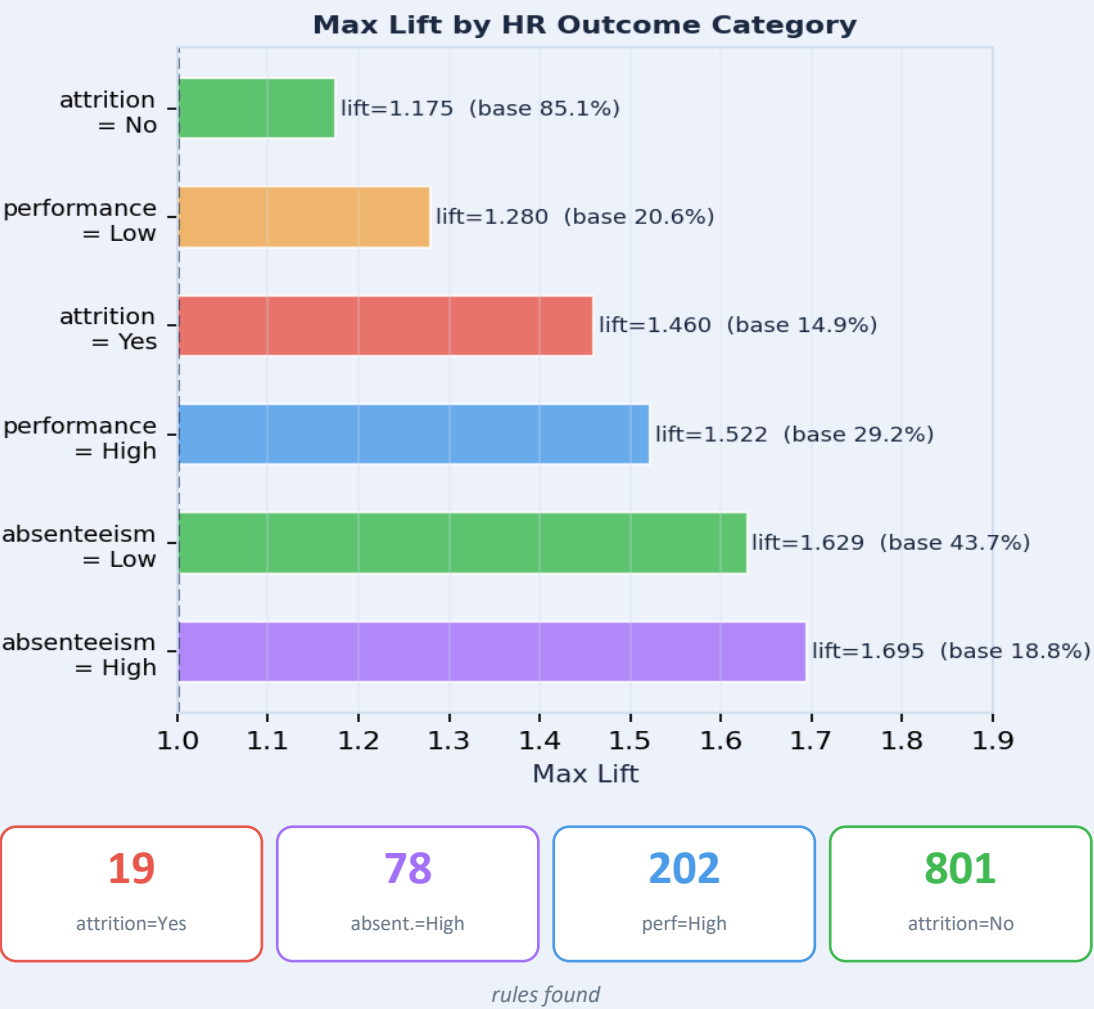
All 11 HR columns are categorical. Every other algorithm required numerical transformation. Apriori treats each (column=value) pair as an item naturally — no encoding assumptions, no information lost to binning or scaling.

Top Rules by Outcome — Ranked by Lift

Outcome	Antecedent Profile	Sup	Conf	Lift
absent. = High	manager_rating=Low · remote_work=No	0.043	0.319	1.665
absent. = Low	remote_work=Yes · shift=Day · training=Completed	0.073	0.712	1.629
performance = High	overtime=No · remote_work=Yes · training=Completed	0.045	0.444	1.500
attrition = Yes	training_status=None	0.044	0.217	1.480
performance = Low	remote_work=No · seniority=Junior	0.048	0.264	1.280
attrition = No ☒	manager_rating=High · remote_work=Yes · training=Completed	0.055	1.000	1.175

3,861 itemsets · 7,561 rules · 2,148 HR-critical · Lift range 1.000 – 1.665

Max Lift by Outcome Category



RISK

Attrition Risk — Highest-Lift Rules

Profile: training_status=None · shift=Night · remote_work=No · manager_rating=Low

Lift 1.480 on attrition=Yes · Confidence 22.4% vs 15.2% baseline



Prioritize training cycle completion. Evaluate night-shift flexibility options. Flag for manager quality review. Do not wait for exit interview — act on profile match.

RISK

High Absenteeism — Strongest Signal in Dataset

Profile: manager_rating=Low · remote_work=No

Lift 1.665 (highest in full ruleset) · Confidence 30.6% vs 18.4% baseline



Investigate management quality in affected teams. This is the single highest-lift HR rule found — 70% above baseline. Poor management + no flexibility = sustained absenteeism. Address the manager, not the employee.

BOOST

High Performance — Evidence-Based Profile

Profile: remote_work=Yes · training_status=Completed · overtime=No

Lift 1.500 on performance=High · Confidence 43.8% vs 29.2% baseline



The highest-performance profile requires no overtime. Flexibility + development + reasonable load = 52% above baseline performance probability. Expand remote access and protect training completion — do not optimize workload at the cost of both.

PROTECT

Retention Lock — 100% Confidence Profile

Profile: manager_rating=High · remote_work=Yes · training_status=Completed

Confidence 100% on attrition=No · Lift 1.175 · Support 5.5% (82 employees)



Zero attrition in this three-condition profile. All three conditions are organizational decisions — not employee characteristics. Invest in manager quality, remote access, and training completion to build retention at scale.

Employee A — High Risk

Department	Operations
Shift	Night
Seniority	Junior
Contract	Temporary
Remote	No
Training	None
Overtime	Yes
Manager	Low

Matching Rules

attrition=Yes · lift 1.480

absenteeism=High · lift 1.665

performance=Low · lift 1.239

🔴 Immediate Action

Employee B — Moderate

Department	Quality
Shift	Day
Seniority	Mid
Contract	Permanent
Remote	No
Training	Partial
Overtime	No
Manager	Medium

Matching Rules

performance=Low · lift 1.128

attrition=Yes · lift 1.089

No absenteeism risk

🟡 Monitor & Develop

Employee C — Retention Profile

Department	Production
Shift	Day
Seniority	Senior
Contract	Permanent
Remote	Yes
Training	Completed
Overtime	No
Manager	High

Matching Rules

attrition=No · conf 100%

performance=High · lift 1.500

absenteeism=Low · lift 1.629

✅ Retain & Promote

Employee A and C work in the same organization, same industry, same compensation band. The difference is entirely structural: training access, remote flexibility, manager quality. The rules found the gap. The policy closes it.

Same organization, three different profiles — three completely different rule matches.

The policy was hiding in the combination.

Not in any individual variable. Not in the average. In the specific co-occurrence of conditions that the organization controls — and can therefore change.

manager_rating=Low + remote_work=No → absenteeism=High · lift 1.665.
Neither condition alone explains the outcome. The combination is the policy gap.

Download. Run it on your own HR dataset.

Lower min_support to find rarer profiles. Raise min_confidence to tighten rules.

**Because every corrective action that cannot name a profile
is just a budget line, not a policy.**

